Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Statistics I

CSIT — 2nd Semester — 2076 — Answer Sheet

Subject Code: STA164

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What are the roles of measure of dispersion in descriptive statistics? Following table gives the frequency distribution of thickness of computer chips (in nanometer) manufactured by two companies.Thickness of computer chips51015202530Number ofchips byCompany A101524201813Company B12182022244Which company may be considered more consistent in terms of thickness of computer chips? Apply appropriate descriptive statistics.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. A study was done to study the effect of ambient temperature on the electric power consumed by a chemical plant. Following table gives the data which were collected from an experimental pilot plant.Temperature(■F)2745725831603474Electric power (BTU)250285320295265298267321(i) Identify which one is response variable, and fit a simple regression line, assuming that the relationship between them is linear.(ii) Interpret the regression coefficient with reference to your problem.(iii) Obtain coefficient of determination, and interpret this.(iv) Based on the fitted model in (a), predict the power consumption for an ambient temperature of 65■F.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. (a) Define Normal distribution. What are the main characteristic of a Normal distribution?(b) What do you mean by probability density function? Write down its properties.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. The following table gives the installation time (in minutes) for hardware on 50 different computers.Installation time0-1010-2020-3030-4040-50TotalNumber of computers4-10-1050If the average installation time is 30.2 minutes, find missing frequencies.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. The length of power failure in minute are recorded in the following table.Power failure time22232425262728Totalfrequency2571043233Find Q1, D2 and P90 and interpret the results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. A manufacturing company employs three analytical plans for the design and development of a particular product. For cost reasons, all three are used at varying times. In facts, plan 1,2 and 3 are used for 30%, 20% and 50% of the products respectively. The defect rate in different procedures is as follows: P(D/P1) = 0.01, P(D/P2) = 0.03, P(D/P3) = 0.02, where P(D/P?) is the probability of a defective product, given plan i. If a random product was observed and found to be defective, which plan was most likely used and thus responsible?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. The random variable X has following probability distribution.X0123456P(X=x)0.030.150.40.20.1.07.05Find (i) E(X) and var(X) (ii) Calculate E(Y) if Y = 3X + 5.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. If two random variables have the joint probability density functionf(x, y) = k(2x + 3y), for 0 = x = 1, 0 = y = 10, otherwiseFind (i) constant k (ii) Conditional probability density function of X given Y (iii) Identify whether X and Y are independent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. A large chain retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 15%. The inspector randomly picks 10 items from a shipment. What is the probability that there will be at least one defective item among these 10?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Messages arrive at an electronic message center at random times, with an average of 9 messages per hour.a) What is the probability of receiving at least four messages during the next hour?b) What is the probability of receiving at most three messages during the next hour?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Following data represent the preference of 10 students studying B.Sc.(CSIT) towards two brands of computers namely DELL and HP.ComputerStudent preferenceLenovo5 2 9 8 1 10 3 4 6 7Acer10 5 1 3 8 6 2 7 9 4Apply appropriate statistical tool to measure whether the brand preference is correlated. Also interpret your result.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. What do you mean by measurement scale? Describe the different types of measurement scales used in statistics.Q.N.13 What is sampling? Discuss various probability sampling techniques with merits and demerits.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
